

An Ensemble of Robust Detectors for Crypto Shock Alerts: A Pre-Registered Study and an Exploratory Follow-Up

SENTINEL project report · October 2026 · Code, protocols and live demo: github.com/nathencoelhoo/sentinel

Abstract. We evaluate SENTINEL, a streaming anomaly detector that combines five robust statistical detectors through a logistic ensemble with per-symbol adaptive alert thresholds, on Binance BTCUSDT and ETHUSDT 1-minute data. **Study 1** (pre-registered, 40 stress days): at a nominal 6 alerts/day SENTINEL emitted fewer alerts than an equally thresholded z-score (4.8 vs 6.7 per quiet day) with no clear change in recall. Because realised alert rates differed and our onset rule was crude, we ran **Study 2** (exploratory, designed after Study 1): 119 new events, a trailing-reference onset rule, and comparison at matched realised alert rates. On 55 held-out shocks at 5 alerts/day, SENTINEL's recall was 0.87 against 0.72 for the z-score (difference +0.15, 95% CI 0.06 to 0.26; the primary endpoint, stated in advance; a random alerter reaches 0.42). The gain was not clear on 58 volatility-regime days (+0.06 [−0.03, 0.15]) and we found no latency advantage. Post-hoc single-detector baselines show the ensemble clearly beats three of its five components, but CUSUM alone and volume-spike alone are within noise of it. The evidence supports an advantage over a naive z-score on large shocks. It does not establish that ensembling is the cause.

1. Introduction

Streaming anomaly detectors for markets are easy to build and hard to evaluate. Volatility clusters, events are rare and hand-labelled, and a detector that alarms often will "catch" almost any event. We built SENTINEL, an open-source streaming detector for crypto markets, and asked: does an ensemble of robust detectors flag large shocks earlier, or with fewer false alarms, than a single z-score? We report two studies. Study 1 was pre-registered; it showed fewer alerts at an equal nominal budget but exposed two flaws in our own design (unequal realised alert rates, a crude onset rule). Study 2 is an exploratory follow-up that fixes both and adds held-out events. Contributions: (i) a causal streaming implementation shared by live and replay use, with 67 automated tests, including one asserting that two engines fed different futures produce identical past outputs; (ii) an evaluation with a random-alert chance baseline, comparison at matched alert rates and cluster-bootstrap intervals; (iii) an account of what is established and what is not, including post-hoc single-detector baselines that limit the main claim.

2. System

Streaming design. Trades are aggregated into 1-minute bars. Every component is a causal state machine that sees bar t only after bars $\leq t$; the same code runs live (in a Web Worker) and in replay. **Detectors.** With r_t the 1-minute log return: (a) *robust z*, $|r_t - \text{med}| / (1.4826 \cdot \text{MAD})$ over the preceding 240 returns [3]; (b) *EWMA volatility ratio* $\sigma_{\text{fast}} / \sigma_{\text{slow}}$ with decay 0.94 and 0.995 [4]; (c) *CUSUM* [2] on standardised returns, $k = 0.5$, restart at $h = 10$; (d) *Bayesian online changepoint detection* [1] (Normal-Gamma, constant hazard 1/200; statistic = posterior mass on run lengths below 5); (e) *volume spike*, v_t over the trailing 240-bar median. An order-book-imbalance detector exists but is unused here (no free historical level-2 data).

Common scale and ensemble. Each raw statistic s_j goes through its own rolling empirical CDF (2,000 observations, queried before insertion) to u_j , then $x_j = -\ln(1 - u_j)$, which is $\text{Exp}(1)$ under stationarity. The ensemble is logistic, $p = \sigma(b + \sum_j w_j x_j)$. *Fixed*: prior $w_j = 1.5/n$, $b = -4.5$. *Learned*: online SGD with an L2 pull toward the prior, non-negative weights, and delayed weak labels (a return of at least $4\hat{\sigma}$ in the next 30 bars, applied only after those bars have elapsed). **Alerting.** The threshold is the $(1 - \rho)$ quantile of the previous 1,440 scores, $\rho = (\text{alarms/day} \times 60 \text{ s}) / 86,400$, with a 15-bar cooldown and a severity-escalation rule that lets a much larger score through inside a cooldown. **Baselines.** *zscore_fixed3*: $|z| > 3$ on a 100-bar rolling mean/standard deviation. *zscore_matched* (the reference): the same statistic with exactly SENTINEL's threshold, cooldown and escalation.

3. Study 1: pre-registered

Protocol. Events: UTC days from 2020-01-01 to 2026-08-31 ranked by $(\text{high} - \text{low}) / \text{open}$; top 20 per symbol, at least 3 days apart (40 events, none excluded). Onset: first minute the close is at least 1% from the UTC day's open. Detection: an alert in $[\text{onset} - 30, \text{onset} + 120]$ minutes. Quiet periods: after a 3-day warm-up, outside every selected event day (60 minutes before to 12 hours after); quiet-period alerts per day are an *upper bound* on false alarms. A random alerter at the same alert rate gives the chance level. Intervals: cluster bootstrap over calendar days (BTC and ETH events on the same day share a cluster), 2,000 resamples [5]; paired differences use identical resamples. The parameters were frozen in one hashed file (SHA-256 prefix c13175749b1c) and committed (f07a4118bb) before event selection ran, a self-hosted registration, not a third-party registry. A five-event pilot replay (COVID crash, Terra/Luna, FTX) preceded it and motivated the design; no parameter was changed afterwards.

Method	Recall [95% CI]	Chance recall [95% range]	Quiet alerts/day [95% CI]	Recall diff vs ref.	Alerts/day diff vs ref.
zscore_fixed3	0.82 [0.69, 0.93]	0.82 [0.70, 0.93]	14.11 [13.38, 14.87]	+0.32 [0.16, 0.50]	+7.46 [6.62, 8.28]
zscore_matched (ref.)	0.50 [0.33, 0.66]	0.53 [0.38, 0.68]	6.65 [6.24, 7.10]	-	-
sentinel_fixed	0.47 [0.30, 0.64]	0.41 [0.28, 0.55]	4.79 [4.52, 5.07]	-0.03 [-0.14, 0.08]	-1.86 [-2.31, -1.42]
sentinel_learned	0.45 [0.28, 0.62]	0.39 [0.25, 0.55]	4.60 [3.99, 5.20]	-0.05 [-0.19, 0.09]	-2.06 [-2.65, -1.45]

Table 1. Study 1 (40 events, nominal budget 6/day). "Quiet alerts/day" is the alert rate outside event periods. **Bold** = paired 95% interval excludes 0.

Findings. (1) At the same nominal budget SENTINEL emitted clearly fewer alerts than the matched z-score, -1.86 per day $[-2.31, -1.42]$. (2) Recall did not clearly change ($-0.03 [-0.14, +0.08]$). (3) The classic $|z| > 3$ rule reached recall 0.82, exactly what a random alerter achieves at its 14.1 alerts/day: high recall bought with alert volume. (4) No clear difference in latency; the learned ensemble was no better than the fixed one; no single detector's removal had a clear effect on recall.

Two flaws in our own design. First, the same nominal budget produced different realised alert rates (4.8 vs 6.7), and a lower alert rate mechanically lowers the chance of a hit, so findings (1) and (2) are not a like-for-like comparison. Second, in 22 of 40 events the onset fell within 30 minutes after 00:00 UTC: these shocks began before the day opened, so the day-open anchor misdated them and added noise for every method. Study 2 addresses both.

4. Study 2: exploratory follow-up

Status. Study 2 was designed after seeing Study 1, to fix its flaws. Only the held-out events are new data for the methods; no detector or ensemble hyperparameter was changed. Its design was fixed in a hashed configuration (prefix 10284e80d82c) before any Study 2 result existed, except that the single-detector baselines (marked *solo*) were added *post hoc* after the main results were seen. All five are reported, not only the best.

- **Events.** Held-out shocks: picks 21 to 50 per symbol under Study 1's ranking rule (60 events). Volatility-regime days: day range at least 4%, ranked by range divided by the median of the previous 7 days, not within a day of any shock day (60 events). One held-out event was excluded for data or onset reasons; 119 were used.
- **Onset.** $\theta = 6 \times$ the median [60-minute log return] over the 7 days *before* the event day; onset = first minute of the day with [60-minute return] $\geq \theta$. Events already above θ at 00:00 UTC (6 of 119) are "in progress" and kept out of the primary analysis (reported as a sensitivity analysis). The multiple 6 (about 4σ for Gaussian noise) was set before any Study 2 result because 4 produced chance crossings on pure noise in a unit test.
- **Matched alert rates.** Every method records a raw score per bar; one alert policy is applied to all of them over a budget sweep (1 to 24 per day), giving each method a recall-versus-realised-alert-rate curve, interpolated at 3, 5 and 8 quiet-period alerts/day. Warm-up is 7 days for every method.
- **Primary endpoint, stated in advance.** Recall difference sentinel_fixed minus zscore_matched at 5 alerts/day on held-out shocks with a fresh onset ($n = 55$); a difference is claimed only if the paired 95% interval excludes 0. Everything else is secondary or descriptive.

Method	Held-out @3/day	Held-out @5/day	Held-out @8/day	Regime @5/day
random alerter (chance)	0.28 [0.16, 0.40]	0.42 [0.29, 0.56]	0.60 [0.45, 0.73]	0.43 [0.31, 0.55]
zscore_matched (ref.)	0.64 [0.50, 0.78]	0.72 [0.60, 0.84]	0.83 [0.73, 0.92]	0.81 [0.71, 0.91]
sentinel_fixed	0.70 [0.58, 0.81]	0.87 [0.78, 0.95]	0.91 [0.82, 0.98]	0.87 [0.76, 0.95]
sentinel_learned	0.70 [0.57, 0.83]	0.81 [0.69, 0.92]	0.91 [0.84, 0.98]	0.83 [0.72, 0.94]
solo: robust_z	0.64 [0.51, 0.78]	0.78 [0.67, 0.89]	0.86 [0.75, 0.96]	0.81 [0.71, 0.91]
solo: ewma_vol	0.53 [0.37, 0.69]	0.74 [0.61, 0.85]	n/a (rate not reached)	0.77 [0.65, 0.89]
solo: cusum	0.70 [0.58, 0.81]	0.81 [0.71, 0.90]	0.90 [0.82, 0.97]	0.85 [0.75, 0.95]
solo: bocpd	0.40 [0.26, 0.54]	0.57 [0.45, 0.70]	0.73 [0.61, 0.84]	0.76 [0.65, 0.87]
solo: volume_spike	0.67 [0.54, 0.79]	0.82 [0.71, 0.91]	0.93 [0.85, 0.98]	0.82 [0.71, 0.92]

Table 2. Recall at a matched realised quiet-period alert rate (held-out shocks, $n = 55$; regime days, $n = 58$), 95% cluster-bootstrap intervals. Chance = a random alerter at the same rate.

Method	Held-out: diff vs z-score	Held-out: diff vs full SENTINEL	Regime: diff vs z-score	Regime: diff vs full SENTINEL
sentinel_fixed	+0.15 [0.06, 0.26] (primary)	-	+0.06 [-0.03, 0.15]	-
sentinel_learned	+0.08 [-0.02, 0.20]	-0.07 [-0.16, 0.03]	+0.02 [-0.08, 0.12]	-0.03 [-0.13, 0.06]
solo: robust_z	+0.06 [-0.04, 0.17]	-0.09 [-0.16, -0.03]	0.00 [-0.11, 0.11]	-0.05 [-0.13, 0.02]
solo: ewma_vol	+0.02 [-0.10, 0.14]	-0.13 [-0.22, -0.06]	-0.04 [-0.18, 0.09]	-0.09 [-0.20, 0.00]
solo: cusum	+0.08 [-0.03, 0.20]	-0.07 [-0.17, 0.03]	+0.04 [-0.06, 0.15]	-0.01 [-0.08, 0.05]
solo: bocpd	-0.15 [-0.28, -0.03]	-0.30 [-0.43, -0.17]	-0.05 [-0.14, 0.05]	-0.11 [-0.22, 0.01]
solo: volume_spike	+0.10 [-0.05, 0.25]	-0.05 [-0.17, 0.06]	+0.01 [-0.10, 0.11]	-0.05 [-0.13, 0.03]

Table 3. Paired recall differences at 5 alerts/day (identical resamples). **Bold** = interval excludes 0. A negative difference vs full SENTINEL means the variant has lower recall than the full ensemble.

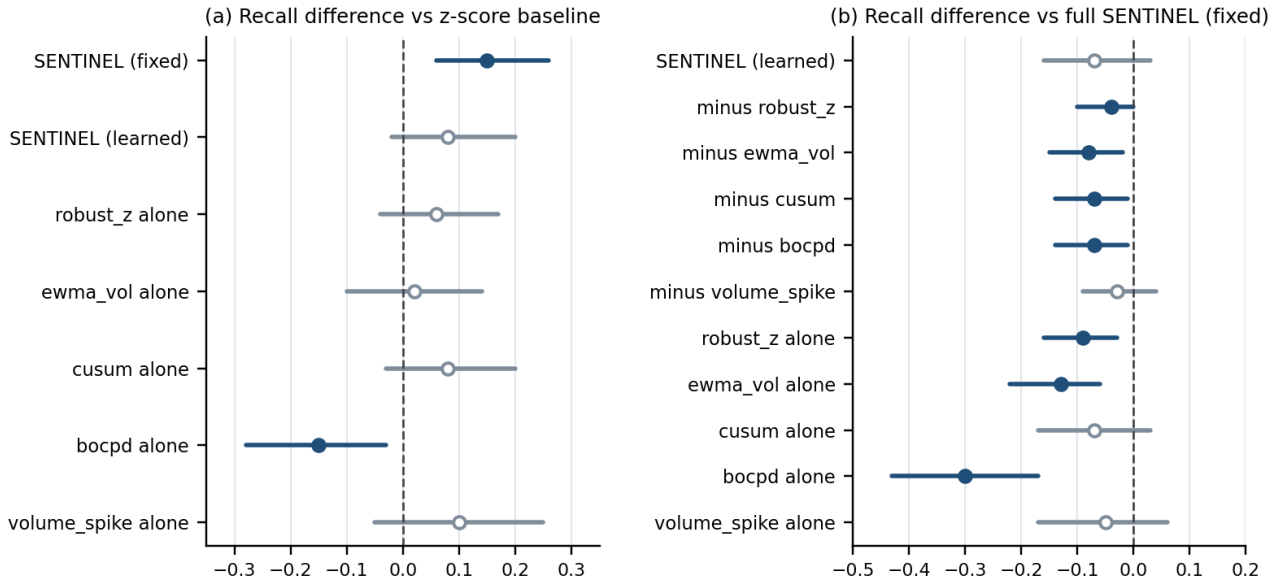


Figure 1. Held-out shocks at 5 alerts/day. (a) vs the z-score baseline; (b) vs full SENTINEL, including leave-one-out and single-detector variants. Filled markers: interval excludes 0.

F1. Primary endpoint met. SENTINEL's recall was 0.87 against 0.72 for the z-score, +0.15 [0.06, 0.26], far above the chance level of 0.42. The direction is the same at 8 alerts/day (+0.08 [0.02, 0.15]) but not clear at 3 (+0.06 [-0.05, 0.16]). At the nominal budget SENTINEL had higher recall (0.87 [0.79, 0.95] vs 0.76 [0.65, 0.88]) with fewer quiet alerts (4.90 vs 5.81 per day). Including in-progress events: +0.14 [0.05, 0.24].

F2. No clear gain on regime days. +0.06 [-0.03, 0.15] at 5/day, +0.09 [-0.01, 0.19] at 3/day, about 0 at 8/day.

F3. No earlier detection. Median latencies lie between -3 and 0 minutes for every method, and every paired latency difference has median 0 (intervals within -1 to +3 minutes) at the 1-minute resolution of the data. **F4. Learned weights do not help** (-0.07 [-0.16, 0.03] vs fixed).

F5. Single detectors (post hoc). The ensemble is clearly better than BOCPD alone (-0.30 [-0.43, -0.17]), EWMA volatility alone (-0.13 [-0.22, -0.06]) and robust z alone (-0.09 [-0.16, -0.03]). CUSUM alone (-0.07 [-0.17, 0.03]) and volume-spike alone (-0.05 [-0.17, 0.06]) are not clearly different from it. BOCPD alone is clearly worse than the z-score baseline (-0.15 [-0.28, -0.03]); no single detector is clearly better than the z-score. **F6. Leave-one-out.** Removing one detector lowers recall by 0.03 to 0.08 at 5/day (the interval excludes 0 for four of five) but not consistently at 3 or 8/day; with this many uncorrected comparisons we treat it as suggestive only.

5. What the evidence supports

Supported.

- At the same nominal alarm budget, SENTINEL raises fewer alerts than an equally thresholded z-score (Study 1; reproduced at the nominal budget in Study 2).
- On held-out large shocks, at a matched realised alert rate, SENTINEL has higher recall than the z-score: +0.15 [0.06, 0.26], a pre-stated primary endpoint.

Not supported or not established.

- Earlier detection (no latency difference); benefit on volatility-regime days; benefit of the online-learned weights.
- That ensembling itself is the cause. CUSUM alone or volume-spike alone is within noise of the full ensemble. Their point estimates are 0.05 to 0.07 lower, so the study can neither exclude an ensemble benefit of that size nor show equality. We did not test mechanisms; the shared normalisation and alert policy are candidates, but robust z alone is also clearly below the full ensemble, which limits that explanation.

- Generalisation to other assets, exchanges or periods; any trading value. No performance claim is made.

6. Limitations

- **Exploratory.** Study 2 was designed after Study 1, the single-detector baselines after the Study 2 results, and the pre-registration is a hashed commit in the author's own repository. Only the primary endpoint carries a pre-stated decision rule; dozens of further comparisons are uncorrected for multiplicity, so a few "clear" secondary differences may be chance.
- **Sample.** BTC and ETH shocks are strongly correlated; the bootstrap resamples calendar days, but the effective sample is small and intervals are wide. Events are the largest-range days, so results concern obvious shocks, not subtle drifts, which Study 2 does not cover.
- **Measurement.** Quiet-period alert rates are upper bounds on false alarms. Matched-rate recall relies on linear interpolation between budget points. The new onset rule (multiple 6) was reasoned, not tuned, and its sensitivity was not tested; recall levels in the two studies are not comparable because the rules differ. Latency resolution is one minute.
- **Design.** One exchange, two assets, 1-minute bars. Each event is replayed with a cold engine and 7 days of warm-up (3 in Study 1). A one-day quantile threshold may raise its own level during multi-day crises; untested. The order-book detector is unevaluated historically. Data are CC BY-NC-SA 4.0 (non-commercial); this is research, not advice.

7. Conclusion and next steps

A robust-detector ensemble with adaptive thresholds gave clearly fewer alerts than an equally thresholded z-score and, on held-out large shocks at a matched alert rate, clearly higher recall. It did not detect earlier, did not clearly help on volatility-regime days, and could not be shown to beat its own best single detectors. A confirmatory Study 3 would fix these in advance: the full ensemble against CUSUM-alone and volume-spike-alone as the named comparators; strictly out-of-time events accumulated after the Study 2 cut-off; additional assets and exchanges; drift events; stronger baselines (for example GARCH-based or isolation-forest detectors); and a labelled set of non-event periods so that precision, not only recall at a matched alert rate, can be measured.

References

- [1] Adams, R. P. and MacKay, D. J. C. (2007). Bayesian online changepoint detection. arXiv:0710.3742.
- [2] Page, E. S. (1954). Continuous inspection schemes. *Biometrika* 41(1/2), 100-115.
- [3] Rousseeuw, P. J. and Croux, C. (1993). Alternatives to the median absolute deviation. *J. Amer. Statist. Assoc.* 88(424), 1273-1283.
- [4] J.P. Morgan/Reuters (1996). *RiskMetrics Technical Document*, 4th ed.
- [5] Efron, B. (1979). Bootstrap methods: another look at the jackknife. *Ann. Statist.* 7(1), 1-26.
- [6] Binance Vision public data, data.binance.vision (licence CC BY-NC-SA 4.0).

Appendix A. Study 1 events

Top-20 daily ranges per symbol, at least 3 days apart. Onset (Study 1 rule) = first minute the close is at least 1% from the UTC day's open. † = onset within 30 minutes after 00:00 UTC (22 of 40). Range % = (high – low)/open. The Study 2 event list is generated by the *study2* workflow (events.v2.json).

BTCUSDT day	Onset (UTC)	Range %	ETHUSDT day	Onset (UTC)	Range %
2020-03-13 †	00:00	45.3	2020-03-13 †	00:00	49.9
2021-05-19 †	00:07	31.7	2021-05-19 †	00:07	46.0
2021-02-08	02:02	22.7	2021-01-03	05:03	31.3
2021-12-04	00:48	22.1	2021-05-23 †	00:02	28.4
2020-03-19	02:04	21.3	2025-02-03 †	00:16	27.7
2021-01-11 †	00:04	20.6	2021-01-11 †	00:04	27.5
2021-01-29 †	00:05	19.8	2020-03-20 †	00:02	26.3
2021-09-07	07:11	19.1	2021-02-23 †	00:05	24.1
2021-05-23 †	00:01	19.1	2021-09-07	07:12	24.0
2022-06-13	00:59	18.7	2022-11-10 †	00:02	23.9
2026-02-06 †	00:05	18.7	2020-08-02 †	00:08	23.5
2022-02-28	01:20	18.0	2025-05-08	01:00	23.0
2020-03-16 †	00:01	17.2	2020-09-05 †	00:22	22.5
2021-02-22 †	00:13	17.2	2022-06-19 †	00:10	22.4
2022-11-08	03:18	17.2	2026-08-19	12:47	22.3
2025-10-10	15:00	16.9	2021-01-21 †	00:15	22.1
2021-05-12	00:37	16.6	2025-10-10	06:52	21.9
2021-01-04	00:31	16.6	2024-08-05 †	00:12	21.8
2023-10-23	01:26	16.2	2024-05-20	04:34	21.0
2021-01-22 †	00:03	16.1	2022-01-21	00:41	21.0

Appendix B. Reproducing the studies

Everything runs on GitHub Actions with no local setup. The *study* workflow reproduces Study 1; the *study2* workflow reproduces Study 2 (select events from Binance Vision daily klines, download the 1-minute files, replay every method, write the tables); *study2b* adds the post-hoc single-detector baselines and reproduces every Study 2 number unchanged (verified on a fixture). Frozen parameters are in eval/study.json and eval/study2.json; protocols are in docs/STUDY.md and docs/STUDY2.md; each result records the configuration hash and commit.